Dual Pathways of Giant Planet Migration & Eccentricity Evolution

First-Principles Dynamical Framework of Dawson & Murray-Clay (2013, ApJ 767:L24)

Pathway A: Metal-Poor Disk ([Fe/H] < 0)

- Lower solid surface density $\Sigma_{\rm solid} \propto 10^{\rm [Fe/H]}$
- Core accretion yields predominantly single giant planets
- Gentle Type II Disk Migration operates smoothly
- Gas corotation & Lindblad resonances damp eccentricity

Outcome: Warm/Cold Giants with Circular Orbits ($e \leq 0.15$)

No Hot Jupiter pile-up; gentle inward stopping

Pathway B: Metal-Rich Disk ([Fe/H] $\geq$ 0)

- High solid surface density exceeds multi-core threshold
- Rapid formation of multiple giant planets ($N \geq 2$–3)
- Disk gas dispersal triggers Violent Planet-Planet Scattering
- Excites Rayleigh eccentricity distribution ($\sigma_e \approx 0.32$)

Outcome: Highly Eccentric Giants ($e \in [0.2, 0.95]$)

Spawns population of migrating Proto-Hot Jupiters

High-e Migration Corridor
$q = a(1-e) \leq 0.08$ **AU**

Stage 3: High-Eccentricity Tidal Circularization & Hot Jupiter Pile-up

- **Orbital Angular Momentum Conserved:** $J = \sqrt{GM_\star\, a(1-e^2)} = \text{const}$
- Viscoelastic tidal dissipation in planet shrinks semi-major axis: $a_{\rm final} = a_0(1-e_0^2) \leq 0.10$ AU
- Rapid circularization timescale $\tau_{\rm circ} \propto (a/R_p)^5 (1-e^2)^{13/2} \ll 1$ Gyr at pericenter

Final Demographics: Prominent 3-day Hot Jupiter pile-up orbiting metal-rich host stars